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Extracted Text

Magnella, Sc.

3. Sulphite of Magnella.

334 Prepris-tion.

1. The compound of fulphurous acid and magnella is formed by pataling fulphurous acid gas into two parts of water, with one of carbonate of magnella. A violent effervescent takes place, with the evolution of heat. The fulphite of magnella is formed, and precipitated to the bottom in the fate of powder / but with an excess of acid it is reaffilatively, and crytallizes.

1553 Properties

2. The crythals of fulphirous magnella are in the form of depredited transparent tetrahedrons. It has a mild earthy tafe, which then becomes family fulphra round; it has no final. Its (pecic gravity is 1) 3/5oz.

3. It effectes in the air, and is flowed converted into fulphate of magenae. It is foluble in 20 parts of cold water. Boiling water didtics a greater proportion, and from this fit crystallizes on cooling. Exposed to heat, this felt becomes vicidé, and by calcination it lofea 0.4; of its weight. If the heat be increased, it is decomposed, the acid is driven off, and the pure earth remains behind.

1354 Achon of air, Sc.

3. It effectes in the air, and is flowed converted into fulphate of magenae. It is soluble in 20 parts of cold water. Boiling water didtics a greater proportion, and from this fit crystallizes on cooling. Exposed to heat, this felt becomes vicidé, and by calcination it lofea 0.4; of its weight. If the heat be increased, it is decomposed, the acid is driven off, and the pure earth remains behind.

The component parts of this fall are,

The component parts of this fall are,

Sulphurous acid 39 Magnella 16 Water 45

1355 Prepras-tion.

1. This triple falls is formed by decomposing the fulphite of ammonia by magnella or the fulphite of magenae by ammonium. In foliation in the cold / or, by mixing together the folutions of the two falls.

1359 Properties

2. This falls is in nonfrentacyrdia, the form of which has had been determined. When it is propelled to the air, it is converted into fulphate of ammonia and magenails. It is leef fulphin in water than either of the two fulphite of which it is formed. By the action of heat, fulphurous acid is given on, aculusful fulphite of ammonia is fublimed, and there remains behind pure

1357 Achon: heat.

5. Nitrate of Magnella.

1358 History and preparation.

1358 History and preparation.

1. This compound of nitric acid and magnella was formerly called _nitre with belfs of magnella_, and magenina _lolpetere_. It is formed by the direct combination of the acid with the earth. By evaporation it is crytallized.

2. This fatty crytlinalizes in four-filled rhomboidal pyrite, whole fuminis are oblique or truncated. Some times it is in the form of small needles combined in groups. The tafte is penetrating and bitter. The pecitic gravity is 1.75.

1359 Properties

3. It is deliquent in the air, and is fsoluble in its own weight of cold water. It is more foluble in boiling water, in which it crytallizes on cooling ; but, it can only be obtained in regular crystals by flow evaporation from its foliation in cold water.

1360 Action of water.

1356 Of heat.

4. By the action of heat it undergoes the waters fufen, the water is driven off, and it becomes dry. It is decompiled in a throat, gives out a little oxygen gas, then nitrous gas, and at all the nitric acid. The pure earth remains behind.

1354 Competi-tion.

The component parts of this fall are, according to

CH E M I S T R Y.

\begin{tabular}{cccc} \thead{\mbox{}} & \thead{Bergman.} & \thead{Kirwan.} & \thead{Magne}\\ Acid & 43 & 46 & \mbox{}\\ Magnella & 27 & 22 & \mbox{}\\ Water & 30 & 32 & \mbox{}\\ 100 & 100 & 100 & \\ \end{tabular}

6. Nitrate of Ammonia and Magnella.

1. This triple falls is formed, either by the direditorgan combination of the folutions of nitrate of ammonia, and nitrate of magnella, by which the falls is obtained pure and crytallized , or, by partially decomposing the nitrate of ammonia by matguela, or the nitrate of magenia by ammonia.

2. The crythals of this fall are and the form of free frozen grains, it has a bitter, acids, and ammonated tafte. It is lefs deliquent in the air than either of the confluent fats, and lefs foluble in water. It requires 11 parts of cold water as differs it, but lefs of boiling water. It crytallizes on cooling.

When it is rapidly heated, it inflamed spontaneously. After when lonly heated in cloke vells, it gives out very heat, gen, gas, and gas, a greater proportion of water than it contains, nitrous gas, and nitric acid, without the familiar trace of ammonia, which flows that it is de-completed, that the hydrogen combines with the oxygen of the acid, and forms water.

The component parts of this fall are,

Nitrate of magnella ammonia 22

100 *

7. Nitrite of Magnella.

Nothing is known of the properties of this fall.

8. Muriate of Magnella.

1. This compound of miruric acid and magnella found was formerly called _mirice, left of magnella_, and was not confounded with the muriate of lime, with which it is frequently recommended. The difference between thefe two falls was flat printed and by Dr Blacks, and Bergman afterwards examined the nature and proper ties of muriate of magnella. The falls is obtained by differing magnella in muricate acid till they are fatuated, and then evaporating the solution. Small irregular crystals are obtained. This falls exists in the waters of the ocean, and in mineral waters, along with the marinate of folds and lime.

2. It is extremely difficult to obtain the muriate of proper magnetic in any regular form. It is either in the late or powder, or very small regular needles or in a kind of jelly. It has a diligenceable bitter tafte. The specific gravity is 1.601.

3. It is very deliquent in the air. Cold water readily dilduces its own weight, and it is fill more lo-ble in boiling water.

4. It is completely decoupled by heat ; the acid is often driven off, and the pure earth remains behind **.*

Let it completely downed by heat ; the acid is often driven off, and the pure earth remains behind **.*

\begin{tabular}{cc} Acid & 34-19\\ Magnella & 41\\ Water & 25\\ \end{tabular}

Source record

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Magnesia, &c.

3. Sulphite of Magnesia.

1352 Preparation. 1. The compound of sulphurous acid and magnesia is formed by passing sulphurous acid gas into two parts of water, with one of carbonate of magnesia. A violent effervescence takes place, with the evolution of heat. The sulphite of magnesia is formed, and precipitated to the bottom in the state of powder; but with an excess of acid it is re-dissolved, and crystallizes.

1353 Properties. 2. The crystals of sulphite of magnesia are in the form of depressed transparent tetrahedrons. It has a mild earthy taste, which soon becomes sensibly sulphureous; it has no smell. Its specific gravity is 1.3802.

1354 Action of air, &c. 3. It effloresces in the air, and is slowly converted into sulphate of magnesia. It is soluble in 20 parts of cold water. Boiling water dissolves a greater proportion, and from this it crystallizes on cooling. Exposed to heat, this salt becomes viscid, and by calcination it loses 0.45 of its weight. If the heat be increased, it is decomposed; the acid is driven off, and the pure earth remains behind.

The component parts of this salt are,

Sulphurous acid	39
Magnesia	16
Water	45
	100

4. Sulphite of Ammonia and Magnesia.

1355 Preparation. 1. This triple salt is formed by decomposing the sulphite of ammonia by magnesia, or the sulphite of magnesia by ammonia, in solution in the cold; or, by mixing together the solutions of the two salts.

1356 Properties. 2. This salt is in transparent crystals, the form of which has not been determined. When it is exposed to the air, it is converted into sulphate of ammonia and magnesia. It is less soluble in water than either of the two sulphites of which it is formed.

1357 Action of heat. By the action of heat, sulphurous acid is given out, acidulous sulphite of ammonia is sublimed, and there remains behind pure magnesia †.

† *Ibid.* p. 85.

5. Nitrate of Magnesia.

1358 History and preparation. 1. This compound of nitric acid and magnesia was formerly called *nitre with base of magnesia*, and magnesian *saltpetre*. It is formed by the direct combination of the acid with the earth. By evaporation it is crystallized.

1359 Properties. 2. This salt crystallizes in four-sided rhomboidal prisms, whose summits are oblique or truncated. Sometimes it is in the form of small needles combined in groups. The taste is penetrating and bitter. The specific gravity is 1.736.

1360 Action of water. 3. It is deliquescent in the air, and is soluble in its own weight of cold water. It is more soluble in boiling water, in which it crystallizes on cooling; but it can only be obtained in regular crystals by slow evaporation from its solution in cold water.

1361 Of heat. 4. By the action of heat it undergoes the watery fusion; the water is driven off, and it becomes dry. It is decomposed in a strong heat, gives out a little oxygen gas, then nitrous gas, and at last the nitric acid. The pure earth remains behind.

1362 Composition. The component parts of this salt are, according to

	Bergman.	Kirwan.
Acid	43	46
Magnesia	27	22
Water	30	32
	100	100

6. Nitrate of Ammonia and Magnesia.

1363 Preparation. 1. This triple salt is formed, either by the direct combination of the solutions of nitrate of ammonia, and nitrate of magnesia, by which the salt is obtained pure and crystallized; or, by partially decomposing the nitrate of ammonia by magnesia, or the nitrate of magnesia by ammonia.

1364 Properties. 2. The crystals of this salt are in the form of fine prisms. It has a bitter, acrid, and ammoniacal taste. It is less deliquescent in the air than either of the constituent salts, and less soluble in water. It requires 11 parts of cold water to dissolve it, but less of boiling water. It crystallizes on cooling.

1365 Action of heat. When it is rapidly heated, it inflames spontaneously. When slowly heated in close vessels, it gives out oxygen gas, azotic gas, a greater proportion of water than it contains, nitrous gas, and nitric acid, without the smallest trace of ammonia; which shows that it is decomposed, that the hydrogen combines with the oxygen of the acid, and forms water.

1366 Composition. The component parts of this salt are,

Nitrate of magnesia	78
——— ammonia	22
	100 *.

* Fourcroy, Comparaison Chim. ii. p. 144.

7. Nitrite of Magnesia.

Nothing is known of the properties of this salt.

8. Muriate of Magnesia.

1367 Found native. 1. This compound of muriatic acid and magnesia was formerly called *marine salt of magnesia*, and was confounded with the muriate of lime, with which it is frequently accompanied. The difference between these two salts was first pointed out by Dr Black, and Bergman afterwards examined the nature and properties of muriate of magnesia. The salt is obtained by dissolving magnesia in muriatic acid till they are saturated, and then evaporating the solution. Small irregular crystals are obtained. This salt exists in the waters of the ocean, and in mineral waters, along with the muriates of soda and lime.

1368 Properties. 2. It is extremely difficult to obtain the muriate of magnesia in any regular form. It is either in the state of powder, or very small regular needles, or in a kind of jelly. It has a disagreeable bitter taste. The specific gravity is 1.601.

3. It is very deliquescent in the air. Cold water readily dissolves its own weight, and it is still more soluble in boiling water.

1369 Action of heat. 4. It is completely decomposed by heat; the acid is driven off, and the pure earth remains behind *.

1370 Composition.

	Bergman.	Kirwan.
Acid	34	34.59
Magnesia	41	31.07
Water	25	34.38
	100	100.04 †.

* Fourcroy, iii. 204.

† Nicholson's Journ. iii. 215.

9. Muriate

Figure 1: *
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